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Virtual Flowmeters and Analyzers are the New Eyes of Smart Plant Operations

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Abstract

Plant operates in dynamic with continuous changes occurring. Although often subtle or seemingly insignificant at normal conditions, nothing in operations remains stagnant. While many non-critical or non-required continuous measurement and monitoring for lines, equipment and devices remain unmeasured, even the main ones can experience measurement accuracy issues, posing challenges to obtain reliable operational data. Stepping up in data accuracy and granularity enhancement, virtual flowmeters and analyzers via metered data-driven process modelling have been established.

The work execution is guided by a structured framework which includes identification of inputs and outputs for the system boundary, mass or gas volume balance establishment, simulation model and user interface development. By leveraging the reliable measurement data from Plant Information (PI) system, the gas balance results for a site facility are robust and credible. The integration of gas balance analysis with process simulation model which can accurately emulate plant operations by determining phase amounts and physical properties through multiphase flash calculations, making it a dependable virtual flowmeter that improves the representation of fluid compositions under actual field conditions.

As part of PETRONAS's commitment in transparent emissions reporting, enhancing the accuracy of methane emissions data is a key priority. This alternative methodology to direct measurement has been demonstrated to fulfil OGMP 2.0 Level 4 requirements in terms of both source-level granularity and high-resolution methane emissions quantification. One of the Upstream Malaysia Asset field operations has shown improved data to represent its low pressure (LP) vent rate with ~19% lower flowrate, indicating reduction in emissions reporting. The integration of continuous updates on process flow rates and gas composition profiles ensures dynamic and accurate emissions estimation under varying operational scenarios. In addition to that, each site facility presents unique challenges and the issues that had been tackled were not common from one site to another. The transparent methodology embedded in the framework

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addresses these challenges by providing a systematic, data-driven approach to validate assumptions, identify missing sources and ensure accurate flow allocation.

Expanding the functionality of the established framework not only enhances the visibility of emissions data but also enables continuous monitoring and verification, thereby significantly increasing the reliability of the site's emissions inventory and footprint. This enables clear visualization of emission pathways, uncovering opportunities for operational optimisation and emissions reduction.

Introduction

Accurate and comprehensive flow measurement is not just critical to operational efficiency, but to environmental stewardship and emissions reporting as well, for oil and gas operations. However, in some upstream facilities, particularly mature or ones located in remote areas, most gas streams remain unmetered or metered by older or less well-maintained instrumentation. These gaps are most prevalent in non-critical streams such as purge lines, gas turbine-compressor casing vents, low-pressure degassing lines, and blanketing gas vents. While these streams are not a focus of process control, they can be substantial contributors to site-level greenhouse gas (GHG) emissions, especially methane.

Industry is increasingly being called to demonstrate precise, open, and high-resolution emissions reporting. This anticipation is triggered by standards such as the Oil and Gas Methane Partnership (OGMP) 2.0, widely regarded as the gold standard for methane reporting. OGMP 2.0 designates five levels of quantification, where Level 4 requires source-level estimation of emissions based on actual site measurement or acceptable engineering-based methods. Specifically, Level 4 demands the measurement of three critical dimensions for each emission source: gas flowrate, methane concentration, and frequency or duration of venting. Only through the combination of the three dimensions can an overall and trustworthy picture of methane emissions be achieved.

To accomplish this, OGMP 2.0 prescribes two general categories of accepted quantification methods:

- **Direct measurement and emission factors** based on measurement that include the application of permanent or temporary equipment (e.g., anemometers, Coriolis meters, thermal mass meters).
- **Alternative methods** based on process simulation, calculations from engineering, and short-term metering campaigns when direct measurement is inappropriate.

In some operating conditions, like PETRONAS upstream asset, direct measurement is not practical or economically viable. Some sites lack physical access to vent points; others have vintage designs in which instrumentation never existed. Even where meters are present, issues related to drift or poor calibration introduce significant uncertainty into reported values. These limitations produce two main challenges:

- Measurement inaccuracies in existing instruments, leading to uncertainty in emissions totals.
- Lack of granularity and allocation accuracy, making it difficult to determine exactly which source is contributing to a reported emission, and increasing the risk of both double-counting and under-reporting.

To overcome these barriers and still meet OGMP Level 4 expectations, this study proposes the use of virtual flowmeters and analyzers. These are data-driven models, built using process simulation software, plant information data (PI system), and validated gas balance calculations, that emulate real-time flow behaviour and thermodynamic properties. By simulating pressure-driven flow across vent systems and calculating phase behaviour using multiphase flash calculations, virtual meters allow for robust estimation of both flowrate and methane content, even in the absence of direct measurement.

Process simulation is uniquely suited to handle the dynamic and complex nature of vent systems, where pressures and compositions vary over time and across operating modes. When coupled with a well-defined

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system boundary and validated against site-wide gas balance, it becomes a powerful tool for source-level emissions quantification. This approach allows operators to meet OGMP Level 4 standards while improving operational visibility and emissions transparency.

The following sections of this paper outline the structured methodology used to develop and implement virtual flowmeters across different venting sources. This includes the classification of vent types, modelling strategies, required input data, and the assumptions used. A field application from an upstream PETRONAS asset is presented to demonstrate the effectiveness and accuracy of this approach in refining methane emissions estimates.

Methodology

This section outlines the structured approach used to develop and apply virtual flowmeters for vent sources within upstream gas processing facilities. The methodology integrates process knowledge, site gas balance data, and rigorous process simulation modelling to quantify methane emissions in accordance with OGMP 2.0 Level 4 requirements. It is divided into three main stages: vent source identification, site gas balance review, and simulation-based flowrate estimation using Aspen HYSYS.

Vent Source Identification

The first step involves identifying all gas venting sources throughout the facility. This is primarily accomplished using Process Flow Diagrams (PFDs) and Process and Instrumentation Diagrams (P&IDs). To facilitate line tracing, engineers typically begin from the vent stack or atmospheric release point and trace backwards to individual vent sources. This reverse-tracing technique ensures complete visibility of both process and utility lines that connect to the vent header.

Each vent source is then categorized based on its flow behaviour into either continuous or intermittent sources, as defined by OGMP 2.0 guidance:

- **Continuous Sources:** These release gas consistently under normal operations. They typically include compressor casing vents, seal gas purge lines, blanketing gas vents (e.g., nitrogen or fuel gas), and low-pressure degassing from vessels. These streams are constantly connected to the vent header without valve actuation and are modelled assuming steady-state pressure-driven flow.
- **Intermittent Sources:** These only vent gas under specific conditions such as overpressure, depressurization, or manual intervention. Examples include Pressure Safety Valves (PSVs), Blowdown Valves (BDVs), Pressure Control Valves (PCVs), Remote Operated Valves (ROVs), Manual Vent Valves, and Pressure/Vacuum Relief Valves (PVRVs). These sources are normally closed and only open during abnormal or infrequent events.

Identifying and classifying vent sources correctly is critical, as the modelling and data requirements differ for each category. A full list of source types, modelling approach, and required data is provided in Table 1.

Site Gas Balance Assessment

To establish a reliable reference for total gas production, consumption, and losses, a comprehensive site gas balance sheet is reviewed. This sheet is typically compiled daily by facility operations and contains well-by-well gas production data, alongside all known gas consumption paths such as fuel gas usage, gas lift, reinjection, and export volumes.

For sites with no metering at vent headers, the venting rate is typically calculated as a balance figure:

Venting (Calculated) = Production – (Export + Other Gas Utilisation (Fuel Gas + Gas Lift +…)

While this method provides a basic estimate, it is inherently uncertain due to the reliance on reconciliation and absence of direct measurement. In some cases, particularly where reconciliation data is inconsistent or unavailable, a simplified estimation is made using a fixed venting ratio. This ratio is developed based

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on historical operating data and applied to the daily export rate to approximate vent flow. Although operationally convenient, this approach introduces further uncertainty and lacks source-level granularity.

These limitations reinforce the need for a more rigorous, source-specific estimation method, which is where simulation comes into play.

Process Simulation Modelling

The virtual flow metering methodology is built within Aspen HYSYS, a process simulation tool capable of performing pressure-driven flow calculations, multiphase flash separation, and compositional tracking. In cases where no existing simulation file is available, a new model is developed using PFDs for major equipment layout and P&IDs for detailed line routing.

Prior to model development phase, engineering drawings are validated with site operations to ensure their accuracy, particularly for vent lines. Many brownfield sites lack updated as-built documentation, so collaboration with platform personnel is necessary to incorporate recent field modifications.

Once confirmed, the base process model is built or updated. This includes verifying key equipment data such as compressor curves, control valve specifications, and BDV/PSV/PCV/MOV datasheets. In most PETRONAS assets, the main process simulation already exists; the focus is therefore on updating:

- **Gas composition**: Obtained from sampling at the wellhead or production separator. For OGMP reporting, samples should reflect the year of reporting and must be saturated (wet gas) since gas chromatography (GC) outputs are typically dry. This ensures accurate methane content estimation in the flash calculation.
- **Operating conditions**: Temperature, pressure, and flowrates from the PI (Plant Information) system are integrated into the model to reflect actual field behaviour.
- **Vent network**: Vent lines are often not modelled in standard process simulations. Therefore, a separate vent network is constructed to simulate gas flow from each source to the LP or HP vent headers and eventually to the atmospheric stack.

For building the vent network, a separate stream is created for each vent source using the "Virtual Stream" block available in the HYSYS model palette. This allows the model to capture essential source conditions—pressure, temperature, and composition—without disrupting the existing process mass balance. Each virtual stream is tapped from the vent release point identified in the P&ID, ensuring that the upstream conditions are accurately represented.

These vent streams are not directly connected to the main process flow. This is intentional: tying them to the main process line would lock the flow due to mass conservation, meaning that changes in valve opening or pressure drop would not affect the simulated flowrate. By keeping the vent streams separate, the simulation can perform pressure-driven flow calculations more effectively and realistically.

The table below summarizes how each type of vent source was modelled in the simulation. It outlines the classification of each source (continuous or intermittent), the general modelling approach applied in HYSYS, and the minimum data required for accurate representation.

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Table 1—Modelling Approaches for Continuous and Intermittent Vent Sources in HYSYS

Source	Type	Modelling Approach	Required Data
Compressor Casing Vent	Continuous	**Pressure-driven flow using pipe segment** Simulate vent line from casing to header. Flow is pressure-driven using upstream casing pressure vs header pressure.	1. Estimated casing pressure (OEM or operation input) 2. Header pressure (PI) 3. Line length, diameter & fittings 4. P&ID
Seal Gas / Purge Line	Continuous	**Pressure-driven flow using dummy control valve.** Flowrate is first estimated from gas balance or OEM data to size the Cv. Valve remains fixed at 100% open in model to simulate purge or seal gas loss. For future use, update upstream pressure & composition only.	1. Estimated flowrate (gas balance or OEM panel) 2. Upstream pressure (PI) 3. Header pressure (PI) 4. Cv (rated or calculated from initial data) 5. P&ID
Blanketing Gas Vent (e.g. N2 purge)	Continuous	**Pressure-driven flow using pipe segment** Model tank blanket gas escaping to header, driven by tank set pressure vs header pressure.	1. Blanket setpoint pressure (PI) 2. Header pressure (PI) 3. Line length, diameter & fittings 4. P&ID
Low-Pressure Equipment Degassing	Continuous	**Pressure-driven flow using pipe segment** Simulate degassing from vessels or drums to header using vessel pressure vs header pressure.	1. Vessel operating pressure/temperature (PI) 2. Header pressure (PI) 3. Line length, diameter & fittings 4. P&ID
PSV (e.g. from separators, compressors)	Intermittent	**Pressure-driven flow using relief valve** Modelled using relief valve block in HYSYS. Opens when upstream pressure exceeds setpoint. Flowrate is calculated based on pressure drop (ΔP) and rated orifice. Valve assumed fully open during overpressure event.	1. Set pressure (P&ID) 2. Full open pressure (OEM/datasheet) 3. Upstream pressure (PI) 4. Header pressure (PI) 5. Orifice designation (D, E, etc.) 6. P&ID
BDV (e.g. from blowdown system)	Intermittent	**Pressure-driven flow using control valve** Blowdown simulated by fully opening control valve (100%). Flowrate is based on pressure differential across valve. On normal days, opening is set to 0%.	1. Vessel/Line pressure & temperature (PI) 2. Valve opening status (0% or 100%) 3. Header pressure (PI) 4. Valve Cv (OEM or estimated) 5. P&ID
ROV / MOV / Manual Vent Valve	Intermittent	**Pressure-driven flow using control valve** Modelled using control valve object. Normally 0% open. When manually activated, set to 100%. Flowrate based on ΔP and valve Cv.	1. Upstream pressure (PI) 2. Header pressure (PI) 3. Valve opening status (from operation) 4. Valve Cv (estimated or from gas balance) 5. P&ID
PVRV (from storage tanks)	Intermittent	**Pressure-driven flow using relief valve** Modelled as relief valve object that opens at tank pressure setpoint. Flowrate based on ΔP between tank and atmosphere. Fully open when active.	1. Set pressure & full open pressure (P&ID / OEM) 2. Tank operating pressure (PI) 3. Vent orifice area or standard size
			4. Backpressure / atmospheric pressure 5. P&ID
PCV (Pressure Control Valve, intermittent use)	Intermittent/ Continuous	**Modelled as control valve with actual valve opening (from PI)** Model PCV using rated Cv and daily valve opening from PI. Flowrate varies depending on opening percentage and upstream pressure.	1. Valve opening % (from PI) 2. Upstream & downstream pressure (from PI) 3. Rated Cv (OEM or datasheet) 4. P&ID

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Before the gas is released into the atmosphere, it typically passes through a knock-out pot or scrubber to remove any entrained liquids. These units are included in the model to represent the final stage of venting and to help determine the actual lean gas composition being emitted. To make this simulation realistic, the knock-out pot is configured with an appropriate backpressure, usually atmospheric pressure plus minor pressure losses from fittings, elevation, and pipe length downstream of the knock-out pot or scrubber.

Accurately modelling this pressure is important as it is not just to reflect real conditions, but also to avoid the risk of air ingress, especially in systems where oxygen could react with hydrocarbons and create a hazardous situation. By applying proper line sizing and realistic pressure drop in the simulation, the model maintains a slight positive pressure throughout the system. This helps ensure one-way flow and reflects good process safety design practice.

For reporting purposes, methane emissions are quantified by extracting two key outputs from the simulation: (1) the vent stream flowrate, measured downstream of the knock-out pot or scrubber, and (2) the methane mole fraction from the flash calculation. Multiplying these values gives the methane mass flowrate for each source, meeting OGMP Level 4 reporting standards.

In summary, this methodology offers a practical and flexible way to estimate vent flow and methane content for sources that are unmetered or difficult to monitor. By combining process simulation with validated field data and sound engineering judgement, the virtual metering framework provides a reliable foundation for OGMP Level 4 reporting. The next section illustrates its real-world application through a field case study.

Results and Discussion

This case study was carried out on an offshore gas production facility within a mature field. The platform operates as a hub, receiving fluids from several bridge-linked satellite platforms, all of which route their vent streams to a shared venting system. From a design standpoint, this means that any gas vented—regardless of source—ends up in a common LP vent header, flows through a knock-out pot (vent scrubber) to remove entrained liquid, and is finally released to the atmosphere.

Overview of Vent Sources and Modelling Scope

A total of 28 venting sources were identified across the facility using the most recent P&ID drawings. These were further validated through direct engagement with the operations team to ensure the sources were still active and correctly reflected in actual field installations. These sources were classified into two main categories:

- **Continuous sources**, such as gas turbine and pump casing vents, fuel gas or nitrogen purge lines, and degassing from closed drain vessels. These streams operate under normal conditions and are assumed to be continuously active unless isolated.
- **Intermittent sources**, which include PSVs, BDVs, ROVs, manual venting lines, and certain PCVs. These only release gas under specific circumstances—whether due to maintenance activities, operational upsets, or emergency shutdowns.

All sources were incorporated into the process simulation using Aspen HYSYS, where each was modelled individually to reflect its operational behaviour. Virtual streams were used at vent tapping points to extract local pressure, temperature, and gas composition—these parameters were then fed into a separate vent network model to simulate flow behaviour. This network, built downstream of the virtual tapping point, included dummy control valves to allow for pressure-driven flow calculations without affecting upstream mass balance.

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Limitations of Previous Estimation Method

Historically, venting flowrates on this platform were estimated using a simplified ratio-based approach. This method relied on applying a fixed venting ratio—determined from previous historical data—to the gas export rate to approximate total monthly venting. For example, if the venting ratio was 0.005 and the export rate was 100 MMscfd, the assumed venting would be 0.5 MMscfd.

This approach became especially relevant from January to July of the reporting year when the LP vent meter on the platform was found to be faulty and awaiting calibration. During this period, the venting values reported in the facility's gas balance sheet were solely based on this ratio method.

While useful as a fallback, this method lacked source-level granularity and could not adjust for real-time operating variation which could lead to significant uncertainty in reported emissions.

Virtual Meter Calibration and Back-Validation

Following the successful calibration of the LP vent meter in August 2020, a simulation model was developed and benchmarked against the now-available metered data. Key operational inputs—including flowrate, pressure, and temperature—were extracted from the plant's PI system and used to mirror actual field conditions within Aspen HYSYS. This ensured that the model accurately captured the behaviour of the vent network.

Once calibrated, the validated model was used to estimate monthly LP venting flowrates for the months preceding calibration (January to July 2020). The model configuration remained fixed throughout this period, while process inputs such as pressure, valve opening, and temperature were updated monthly based on historical PI data. All vent sources used in the model were previously confirmed with operations, and their configurations were assumed unchanged throughout the assessment period.

The results revealed a consistent trend: from January to July, the simulation-based estimates were lower than the previously reported values calculated using the fixed-ratio method. On average, the difference between the two methods amounted to an overestimation of approximately overall 19% in the original fixed-ratio approach.

As illustrated in Figure 2, there is a clear divergence between the two estimation curves across the pre-calibration period. The fixed-ratio approach, represented by the purple line, remains consistently higher than the blue line, which represents the simulation-based estimates. This reflects the inherent limitations of using a static ratio—typically applied as a fixed percentage of the export gas flowrate—without accounting for real changes in venting conditions or source behaviour.

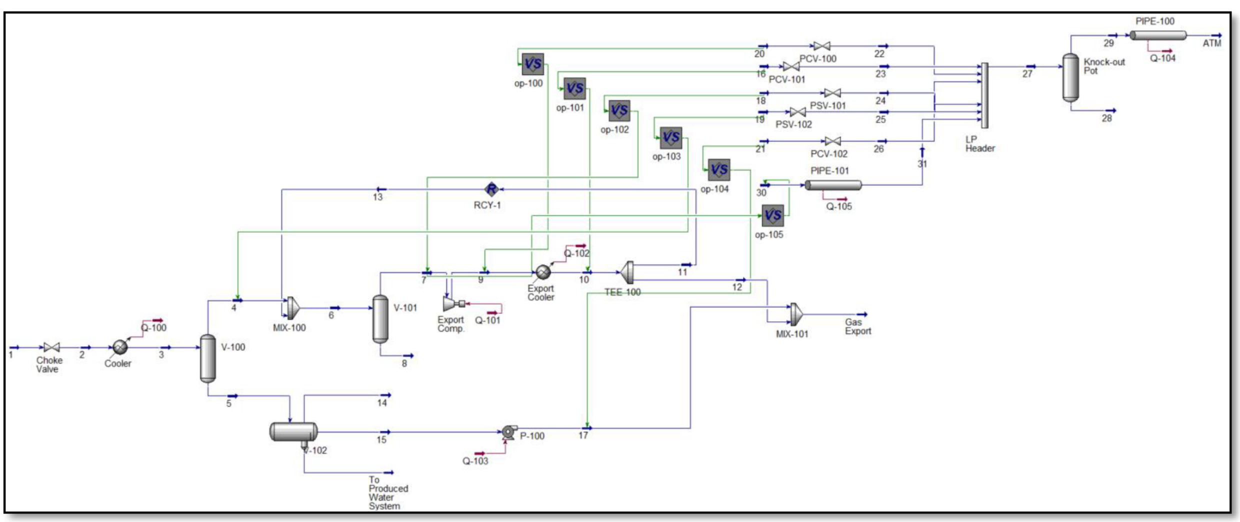

Figure 1—Overview of vent network at one of the satellite platforms

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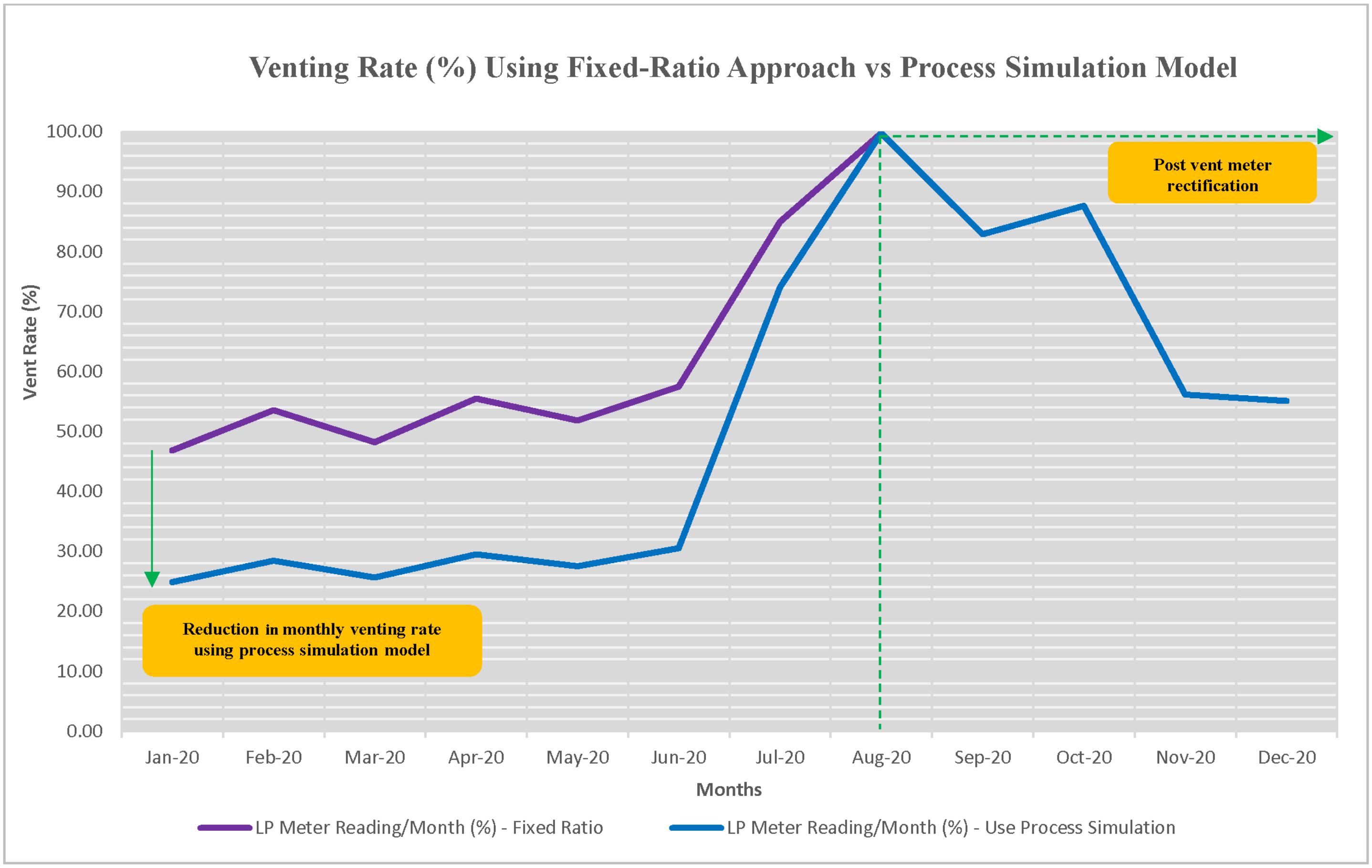

Figure 2—Venting trends using different approaches

In contrast, the simulation model incorporates source-level detail, including actual upstream and downstream pressures, valve Cv values, real-time valve openings (where available), and header backpressure. By simulating flow as pressure-driven through realistic flow paths, the model produces estimates that are more aligned with actual operating behaviour.

From August 2020 onward, following the meter's rectification, the fixed-ratio method was no longer used. The monthly reporting was based on calibrated meter data, and the simulation was used solely for validation purposes. The convergence of the simulation output with the metered data post-August further confirms the reliability of the model as a virtual metering tool, especially in scenarios where physical instrumentation is not available or temporarily compromised.

Several key insights emerged from the simulation and back-validation process:

- **Improved granularity:** The simulation differentiates between continuous and intermittent vent sources, treating each with a tailored modelling approach. This granularity enables more accurate tracking of real venting activity, even in complex systems with multiple contributors.
- **Pressure differentials matter**: Unlike the fixed-ratio method, which assumes constant proportionality, the simulation responds to real plant dynamics—calculating flow based on actual pressure drops across vent valves and equipment.
- **Valve-specific behaviour is captured**: For example, pressure control valves (PCVs) were modelled using actual valve openings from the PI system and corresponding Cv values from datasheets. This allowed flowrates to reflect partial openings and throttling, rather than relying on fixed binary assumptions.

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- **Realistic handling of intermittent events**: Intermittent sources, such as BDVs and PSVs, were only switched to open in the model when a venting event was known or expected. This provided a more faithful representation of actual release behaviour, avoiding unnecessary overreporting.

The 19% gap between the fixed-ratio method and the simulation-based estimates is non-negligible, particularly for methane and GHG emissions reporting. Overstating venting flowrates not only affect performance tracking but can also lead to misaligned environmental disclosures and regulatory concerns.

By integrating process simulation into the venting estimation workflow, several benefits are realized:

- More accurate and transparent emission inventories, which are critical for environmental compliance and initiatives such as OGMP 2.0.
- Improved internal reporting confidence, enabling operators to focus on real issues rather than misrepresented data.
- Enhanced operational insight, particularly during periods where instrumentation may be offline or unreliable.

Ultimately, this study demonstrates that process simulation—when calibrated and supported by actual plant data—can serve as an effective virtual meter. Not only does it fill the gaps left by faulty instrumentation, but it also establishes a more robust foundation for emission estimation and reporting moving forward.

Conclusion

This study demonstrated the value of using process simulation as a virtual metering tool to estimate gas venting, particularly in cases where physical instrumentation is unavailable or unreliable. By modelling each vent source based on its physical configuration, pressure differential, valve behaviour, and process data, a more representative and granular picture of actual venting activity was achieved.

The validation exercise—comparing simulation output with calibrated meter readings—confirmed the robustness of the modelling approach. Notably, the simulation consistently reported lower and more realistic venting volumes compared to the traditional fixed-ratio method. Across a seven-month period prior to meter rectification, the fixed-ratio approach was shown to overreport venting by an average of 19%, underscoring the limitations of generalised estimation techniques when applied across diverse and dynamic venting sources.

The simulation approach allowed for differentiation between continuous and intermittent sources, incorporated real-time valve openings where applicable, and respected the actual operating pressures across each flow path. This not only improved reporting accuracy but also provided deeper insights into how venting events correlate with equipment operation and plant conditions.

Importantly, the model's ability to replicate the post-calibration metered values adds confidence to its application as a virtual meter—one that can be used for daily, monthly, or event-based reporting. This becomes especially critical in offshore operations where metering redundancy is limited, and emission reporting frameworks such as OGMP 2.0 demand a higher standard of traceability and accuracy.

In summary, process simulation offers a practical and technically sound solution for vent estimation. It bridges the gap between engineering understanding and regulatory expectations, enhancing both operational awareness and environmental accountability. Moving forward, such models can be integrated into regular reporting workflows—not only as a backup tool, but as a standardized part of emissions quantification.

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